



6th International Innovation Competition (ITC-EGYPT 2026) Graduation Projects' Template

Faculty/school: Faculty of Computers and Data Science

University / Studying administration: Alexandria University

University Logo:



Project Title:

Alex Flood Risk Prediction

Supervisor:

Name	Email	WhatsApp numbers	position
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Team Members:

NO	Name	Email	WhatsApp numbers
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Executive Summary

- Alexandria Flood Risk Prediction project is a data-driven system designed to deliver highly precise, location-based flood risk assessments across Alexandria's coastal areas. By integrating environmental, oceanographic, and demographic data, the system accurately identifies which areas are at risk and determines the severity of their exposure - whether from heavy rainfall or the Mediterranean Sea.
- The project addresses a critical and growing problem: the lack of accurate, fine-grained flood risk analysis in a city increasingly threatened by climate change, sea level rise, and extreme weather. Unlike generalized or speculative approaches, this system enables highly precise, location-level identification of vulnerable areas. By feeding real-time or forecasted environmental measurements into the model, it can generate a high-resolution grid visualization across Alexandria, showing the distribution of flood risk for each cell at a specific time - whether driven by rainfall or sea conditions - providing actionable, data-backed insights for real-world decision-making.
- Practically, the project is highly feasible. It relies on globally available datasets and established data science technologies, making it implementable without requiring expensive physical infrastructure. This allows for rapid deployment and scalability.
- Currently, the project is being developed independently without partners or external funding, with potential for future collaboration with government entities, insurance companies, or investors interested in environmental risk management.
- The value it adds to industry is significant. It can transform how governments plan coastal protection, how insurance companies assess and price risk, and how investors evaluate real estate safety. Instead of reactive decisions after disasters, this project enables proactive, data-driven planning.
- Importantly, the system can be developed into a real industrial product - either as a decision-support platform, an interactive risk-mapping system, or an API service delivering flood risk scores. This creates strong potential for commercialization and integration into existing infrastructure and planning systems.
- Scientifically, the project is grounded in machine learning, spatial analysis (GIS), time-series modeling, and physics-informed risk estimation, using trusted global data sources such as NOAA, ECMWF, NASA, and Copernicus.

- The system is primarily software-based, utilizing Python, machine learning libraries, and cloud computing platforms. It does not require physical hardware, making it cost-efficient and accessible.
- Expected beneficiaries include government agencies, urban planners, insurance companies, real estate developers, and environmental researchers—any stakeholder needing accurate flood risk insights.
- The project also presents a strong opportunity for building a startup, offering services through SaaS platforms or APIs. Initial costs are relatively low due to open data usage, with a clear business model based on subscriptions or enterprise partnerships.
- The current cost of the project is relatively low, relying mainly on open-source data and free analytical tools.
A proposed business model includes offering the system as:
 1. A decision-support tool for government authorities
 2. A paid service or API for insurance companies
 3. An analytical tool for real estate investors and developers
- Future development includes adding more features like: integrating real-time data, expanding to other coastal cities in Egypt, and developing early warning systems.
- In summary, this project is not just a technical solution - it is a high-impact tool capable of improving safety, reducing economic losses, and enabling smarter, data-driven decisions in the face of increasing climate risks.

Team Structure and Responsibilities

Name	Contribution (%) and Description
Mustafa Younis	Contact Information: phone & WhatsApp: 01273522694 email: mostafayounis1229@gmail.com Contribution(100%): -Sole developer and project owner. Responsible for the entire Project including: - data collection - data preprocessing - data wrangling - feature engineering - model development - evaluation - visualization and overall system design and implementation

Partners and financiers.

- Currently, the project is being developed independently without any partners or external financiers. However, there is strong potential for future partnerships or funding from government entities, insurance companies, or investors interested in environmental risk management and smart city solutions.

Abstract

- **The problem, the proposed approach, and my solution.**

Coastal flooding has become an increasing concern for cities like Alexandria due to rising sea levels, extreme weather conditions, and urban expansion. While the existence of flood risk is widely recognized, there is a lack of precise understanding of its exact geographic distribution across specific locations. Much of the current discussion is based on assumptions rather than data-driven analysis. This project addresses the need for an accurate and dynamic system to assess and predict flood risk at a fine spatial and temporal scale.

The proposed approach is based on integrating multiple environmental, oceanographic, and demographic factors such as sea level, weather conditions, wave characteristics, elevation, and population density. Alexandria's coastline is divided into small grid-based coastal cells, allowing localized analysis. Machine learning models are trained on historical data (from around 2000 onwards) combined with simulation-based and domain-driven risk estimation techniques.

- **Main contributions of my work**

The core contribution of this project is its ability to take real-time or future input measurements of these factors - at any specific time (with hourly precision)—and instantly predict the distribution of flood risk across all coastal cells along Alexandria. The system outputs a normalized Flood Risk Score for each cell and presents results through an interactive dashboard, enabling clear visualization of high-risk and safe areas.

- **Key results and conclusions.**

Key results demonstrate that combining data-driven modeling with environmental understanding provides more reliable and comprehensive flood risk predictions. The project concludes that such a system can serve as a powerful decision-support tool for governments, urban planners, and investors, enabling proactive risk management, better planning decisions, and improved resilience against coastal flooding.

Abbreviations and Terms

Abbreviation	Definition
Flood Risk Score (FRS)	A normalized value (0–1) representing the likelihood or severity of flooding in a specific coastal cell.
Coastal Cell (Grid Cell)	A small spatial unit (e.g., 500m × 500m) used to divide the study area for localized analysis.
Machine Learning (ML)	Algorithms used to identify patterns in data and predict flood risk based on multiple input factors.
EDA (Exploratory Data Analysis)	The process of analyzing and summarizing datasets to understand their structure and patterns.
API (Application Programming Interface)	A service interface that allows external systems (e.g., insurance platforms) to access the model's predictions.
Bathymetry	The study of underwater depth and seabed shape, influencing wave behavior and flooding.
DEM (Digital Elevation Model)	A dataset representing land elevation used to assess flood vulnerability.
Storm Surge	A temporary rise in sea level caused by storms, often leading to coastal flooding.
Tidal Range	The difference between high tide and low tide levels.
Wave Run-up	The maximum vertical extent waves reach on the shore above sea level

Introduction

- **What is the problem?**

Coastal flooding is a growing concern for Alexandria due to rising sea levels, climate change, and increasing urbanization along the coast. While the existence of flood risk is widely recognized, there remains a critical gap in identifying which specific areas are most vulnerable and how flood risk is distributed geographically across different locations. Current approaches are often too generalized and fail to provide high-resolution, location-specific insights.

- **Why is it important?**

This problem is important because understanding the **exact spatial distribution of flood risk** is essential for accurate decision-making. Without precise mapping of high-risk and low-risk zones, stakeholders cannot effectively assess impact at the local level, whether for planning, investment, or risk evaluation.

- **Why hasn't it been solved before?**

Because the main challenge lies in the complexity of combining multiple dynamic factors-such as sea level, weather conditions, wave behavior, and topography-into a unified system that produces detailed, location-based predictions. Additionally, the absence of complete historical flood data makes it difficult to rely solely on traditional modeling techniques.

- **How does your solution differ?**

This project introduces a data-driven approach that focuses on high-resolution flood risk distribution rather than general prediction. By dividing Alexandria's coastline into fine-grained spatial grid cells and integrating multi-source data, the system can estimate a Flood Risk Score for each location. A key advantage of this approach is its ability to take input measurements at any specific time (with hourly precision) and generate a detailed map showing how flood risk varies across all coastal areas.

- **What are the key components of your approach?**

- **Fine-grained spatial grid system:** Divides Alexandria into small coastal cells to provide detailed, location-specific analysis instead of general city-level risk.
- **Time-specific prediction inputs:** Uses environmental conditions at a specific timestamp (hourly precision) to capture how risk changes over time.
- **Spatial flood risk mapping:** Produces a separate Flood Risk Score for each coastal cell, showing how risk is distributed across different areas.
- **Hybrid risk modeling:** Combines historical data, simulation-based estimation, and a weighted environmental formula to improve accuracy even with limited real flood labels.
- **Multi-source feature integration:** Merges oceanographic, weather, elevation, and demographic data into a unified model for better risk understanding.
- **Interactive dashboard output:** Visualizes flood risk distribution across Alexandria, enabling clear interpretation of high and low-risk zones.

Key difference:

The system focuses on **detailed spatial distribution of flood risk**, not just overall prediction, enabling precise identification of affected areas.

Novelty

The main novelty of this project lies in shifting flood risk analysis from a general, city-level prediction to a **high-resolution spatial and time-specific (hourly) risk distribution system**. Unlike existing solutions that typically provide overall flood risk estimates, this approach generates a **detailed Flood Risk Score for each individual coastal cell in Alexandria**, showing exactly where and how strongly each area is affected.

Another innovative aspect is the ability to take environmental measurements at a **specific point in time with hourly precision** and instantly map how risk is distributed across all coastal locations. In addition, the project introduces a **hybrid risk modeling framework** that combines historical data, physics-based simulation, and a weighted environmental formula, allowing reliable predictions even in the absence of complete historical flood records.

This differentiation is valuable because it transforms flood analysis into a **localized, actionable, and dynamic system**, enabling much more precise understanding of vulnerable areas rather than broad assumptions.

Methodology

- **Detailed architecture of the proposed project.**

- The project follows a standard data science pipeline adapted to a geospatial flood risk system. It starts with data collection from multiple sources (e.g., NOAA, ECMWF, Copernicus), followed by data cleaning, integration, feature engineering and modeling. All features are strictly linked to geographic coordinates (latitude & longitude), ensuring full spatial consistency across datasets.
- Features are divided into two main types: **dynamic features**, which are time-dependent and derived from historical or real-time data (such as weather conditions, sea level, waves, and storm surge), and **static features**, which are relatively constant and taken as latest standard measurements (such as elevation, bathymetry, soil type, and coastal structure).
- The entire study area of Alexandria is then transformed into a **fine-grained spatial grid (coastal cells)** with very small cell sizes to maximize resolution and prediction accuracy. Each cell aggregates all relevant features based on its latitude and longitude position.
- After preprocessing, a **hybrid modeling approach** is applied, combining historical data, simulation-based estimation, and weighted environmental factors to compute a Flood Risk Score for each cell.

- **The technology used.**

- The project uses Python as the main technology, with Pandas, NumPy, Scikit-learn, and GeoPandas for data processing, machine learning, and geospatial analysis.
- Web scraping is handled using tools like BeautifulSoup and Requests to collect external data.
- while NetCDF datasets are processed using xarray and netCDF4 for handling large-scale climate and oceanographic data.
- SQL databases are used for structured data storage and querying.
- For visualization and deployment, tools such as Power BI
- web-based frameworks like Streamlit/Plotly Dash are used to build an interactive dashboard that displays flood risk distribution across Alexandria's coastal grid.

- **Any special cases (if any).**

- The project addresses missing or incomplete historical flood data using a hybrid labeling approach that combines simulation-based estimation and a weighted environmental risk formula.
- Static features (such as elevation and bathymetry) are treated using latest standard measurements, while dynamic features are time-dependent and updated at hourly resolution.
- Extreme weather events are handled as special high-impact cases to improve model robustness.
- In addition, a key challenge is the **difference in spatial and temporal resolution across datasets**, which is resolved by aligning all variables to a unified fine-grained grid structure and standardizing them to a common spatial-temporal resolution before modeling.

- **Error analysis (if any).**

- Model performance is evaluated using standard error metrics such as MAE, RMSE, and spatial consistency checks to ensure reliable Flood Risk Score predictions across all grid cells. Special attention is given to errors caused by data imbalance, especially in rare extreme flood events, as well as uncertainties arising from combining datasets with different resolutions and missing historical labels. Cross-validation and spatial validation techniques are used to assess generalization across different coastal regions, ensuring the model remains stable and accurate under varying environmental conditions.

Results

- The implementation of the proposed methodology successfully produces a high-resolution, data-driven system for analyzing flood risk distribution across Alexandria's coastline. The model achieves the project's main objective by generating a **Flood Risk Score for each coastal grid cell in specific time unit like hour**, allowing precise identification of how flood risk varies spatially across different areas rather than providing a general city-wide estimate.
- One of the key outcomes is the ability to process both static and dynamic environmental factors, aligned to latitude and longitude, and produce consistent risk maps at different time instances with hourly precision. The system effectively integrates multi-source datasets and resolves differences in spatial and temporal resolution, resulting in reliable and unified risk estimation.
- The final output is presented through an **interactive dashboard**, which visualizes flood risk distribution across Alexandria in an intuitive and spatial format. Users can explore different time scenarios and immediately observe how risk levels change across coastal cells.
- Additionally, the system demonstrates the potential to be extended into a **real-time monitoring and early warning tool**, where high-risk zones can be highlighted and alerts can be generated based on risk thresholds. This enhances its practical value for decision-making in urban planning, disaster preparedness, and risk management.
- Overall, the results confirm that the proposed approach successfully achieves fine-grained, time-sensitive flood risk mapping, providing a more detailed and actionable understanding of coastal vulnerability compared to traditional generalized models.

Product

Data Science system to predict flood risk in Alexandria, generate risk maps, and accurately identify future high-risk areas with high precision to support proactive decision-making.

- **Applicability.**

- Coastal cities
- Urban planning
- Insurance
- Real estate

- **Features.**

- Grid-based analysis
- Risk score (0 → 1)
- Interactive maps
- Multi-data analysis (weather, sea, population)
- API access

- **List the electronic, physical, and software components used in the project.**

- **Software:** Python, ML, GIS
- **Electronic:** Cloud servers
- **Physical:** Data centers

- **Cost of the prototype.**

- 5,000 – 10,000 EGP

- **Expected cost of the product.**

- 20,000 – 40,000 EGP

- **Market size.**

- Large global demand (billions), especially governments & insurance.

- **Expected customers / Early customer interest.**

- Governments
- Insurance companies
- Real estate developers

- **Business Model Canvas.**

- SaaS subscription
- Paid API
- Custom reports

- **Top 3 competitors.**

- ClimateAI
- JBA Risk
- Fathom

- **Pricing strategy.**

- Tiered (Basic / Pro / Enterprise)

- **Risk assessment and mitigation.**

- Data quality
- Model accuracy

Mitigation: multiple data sources + continuous improvement

- **Scaling strategy.**

- Start in Alexandria → expand along Egypt

References

1. Climate & Weather Data

- European Centre for Medium-Range Weather Forecasts (ECMWF) – ERA5 Reanalysis Dataset
<https://www.ecmwf.int/en/forecasts/dataset/ecmwf-reanalysis-v5-era5>
 - Copernicus Climate Data Store (CDS)
<https://cds.climate.copernicus.eu>
 - National Oceanic and Atmospheric Administration (NOAA)
<https://www.noaa.gov>
 - OpenWeather API
<https://openweathermap.org/api>
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2. Sea Level & Ocean Data

- Copernicus Marine Environment Monitoring Service (CMEMS)
<https://marine.copernicus.eu>
 - Permanent Service for Mean Sea Level (PSMSL)
<https://www.psmsl.org>
 - NASA Jet Propulsion Laboratory (JPL) Sea Level Data
<https://sealevel.jpl.nasa.gov>
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3. Wave & Ocean Dynamics

- ECMWF ERA5 Wave Reanalysis
 - NOAA National Data Buoy Center (NDBC)
<https://www.ndbc.noaa.gov>
 - HYCOM (Hybrid Coordinate Ocean Model)
<https://www.hycom.org>
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4. Bathymetry (Seafloor Depth)

- GEBCO (General Bathymetric Chart of the Oceans)
<https://www.gebco.net>
 - EMODnet Bathymetry
<https://emodnet.ec.europa.eu>
 - International Bathymetric Chart of the Mediterranean (IBCM)
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5. Coastal & Land Data

- ESA WorldCover 2023
<https://esa-worldcover.org>
 - Copernicus Global Land Cover
<https://land.copernicus.eu>
 - MODIS Land Cover Dataset (NASA)
<https://modis.gsfc.nasa.gov>
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6. Elevation Data

- Shuttle Radar Topography Mission (SRTM) – NASA/USGS
<https://www2.jpl.nasa.gov/srtm>
 - ASTER Global Digital Elevation Model (ASTER GDEM)
<https://asterweb.jpl.nasa.gov>
 - Copernicus DEM
<https://spacedata.copernicus.eu>
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7. Soil & Ground Data

- ISRIC SoilGrids (World Soil Information)
<https://soilgrids.org>
 - Harmonized World Soil Database (HWSD) – FAO/IIASA
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8. Population Data

- WorldPop Project
<https://www.worldpop.org>
 - GPWv4 (Gridded Population of the World) – SEDAC NASA
<https://sedac.ciesin.columbia.edu>
 - HRSL (High Resolution Settlement Layer) – Facebook/Meta Data for Good
<https://dataforgood.facebook.com>
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9. Tide & Storm Surge Data

- NOAA Tides & Currents (CO-OPS)
<https://tidesandcurrents.noaa.gov>
- Global Extreme Sea Level Analysis (GESLA)
<https://gesla.org>

- SURGEDAT Global Storm Surge Database
 - Global Tide and Surge Model (GTSM) – ECMWF/ERA5-based
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10. Coastal & Mapping Data

- OpenStreetMap (OSM)
<https://www.openstreetmap.org>
 - Sentinel-2 Satellite Data (ESA Copernicus)
<https://sentinel.esa.int>
 - S2Coast Dataset (Google Earth Engine Community)
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11. Groundwater Data

- International Groundwater Resources Assessment Centre (IGRAC)
<https://www.un-igrac.org>
 - Global Groundwater Monitoring Network (GGMN)
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12. Related Scientific & Industry Sources

- ClimateAI (Climate Risk Analytics Platform)
<https://www.climate.ai>
 - JBA Risk Management
<https://www.jbarisk.com>
 - Fathom Global Flood Risk Data
<https://www.fathom.global>
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Summary

All datasets used in this project are:

- Open-source or scientific research datasets
- From reputable organizations (NASA, NOAA, Copernicus, ESA)
- Suitable for environmental and flood risk modeling